Week #10: [Foot & Ankle]

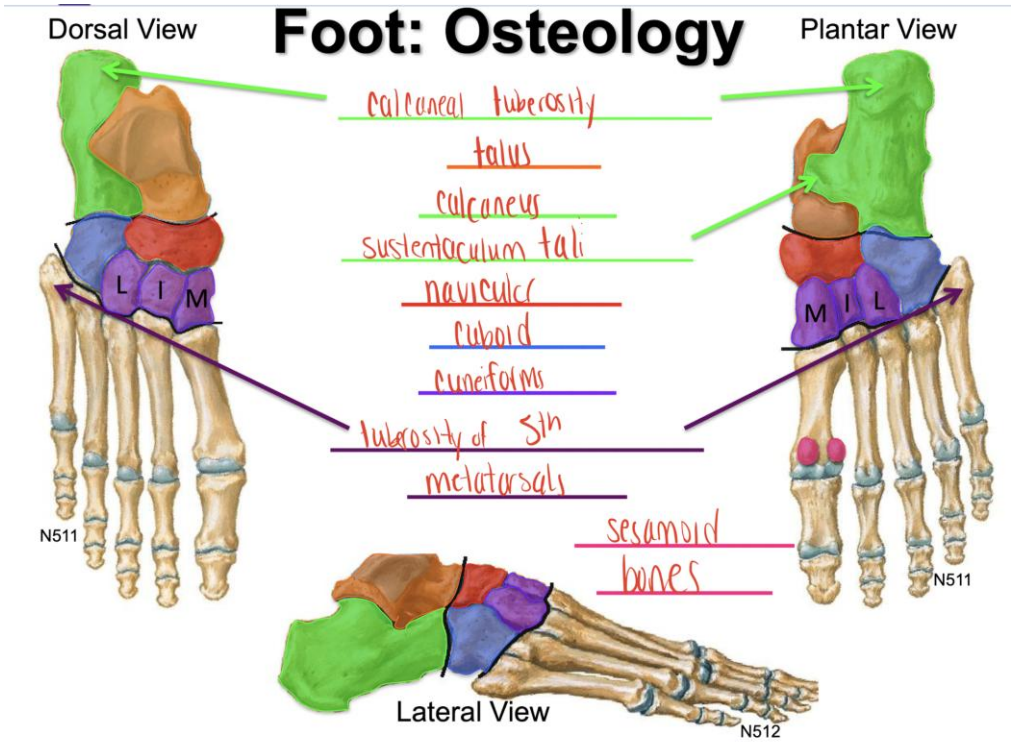
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General Organization of the Foot

- Review Osteology of the Foot
- Soft Tissues of the Dorsum and Plantar Aspect of the Foot





Tibia – The Tibia is the major weight bearing bone of the leg. It extends superficially into the subcutaneous tissue in the anteromedial leg earning its nickname the "shin bone." Because of its superficial location, the periosteum of the tibia is very susceptible to bruising.

a. **Medial Malleolus** – widening of the tibia distally for articulation with and weight transfer to the Talus.

Fibula – The fibula lies posterior and lateral to the tibia. The **head of the fibula** articulates with the tibia proximally and the **lateral malleolus** articulates with the talus

distally to help hold the talus in place with the medial malleolus of the tibia. The fibula does not have a weight bearing function, and for this reason, is the bone of choice in bone grafts (removal of part of the fibula does not necessarily take away any ability to walk, run or jump)

OSTEOLOGY

A. Seven Tarsals:

1. **Talus** – The talus is held in place by the medial and lateral malleoli of the tibia and fibula respectively. It articulates with the calcaneus inferiorly and the navicular anteriorly. Its function is to transfer weight from the tibia to both the calcaneus and forefoot via articulation and the forefoot via ligaments.
2. **Calcaneus** – The calcaneus forms the heel and is the largest bone of the foot. It transfers weight from the talus to the ground. It articulates with the cuboid and the talus. The **calcaneal tuberosity**, located on the posterior surface of the calcaneus, is the insertion site for the calcaneal tendon ("Achilles Tendon"); and the **sustentaculum tali**, located superomedially, provides a shelf for the talus to sit on as well as a pulley for the flexor hallucis longus tendon to pass under.
3. **Navicular** – A boat shaped tarsal that articulates with the talus posteriorly and the three cuneiform bones anteriorly. The navicular tuberosity, a prominence on the inferior aspect of the navicular, provides a key attachment site of the calcaneonavicular ligament ("Spring Ligament"), which helps to maintain the longitudinal arch along the medial foot.
4. **Cuboid** – the most lateral of the tarsal bones. The cuboid articulates with the calcaneus, talus, lateral cuneiform, navicular, and 4th and 5th metatarsals.
5. **Medial, Intermediate, and Lateral Cuneiform (First, Second and Third Cuneiform)** – Articulate with the navicular and cuboid posteriorly and the first four metatarsals anteriorly. Their wedge shape contributes greatly to the formation and maintenance of the transverse arch of the foot.
6. **Metatarsals and Phalanges** – Provide attachment sites for a number of muscles that move the foot and its digits. The **tuberosity of the 5th metatarsal** extends posteriorly off the 5th metatarsal and serves as an insertion site for the fibularis brevis. When the foot is inverted quickly, the fibularis brevis can pull the tuberosity of the 5th metatarsal off (avulsion fracture).

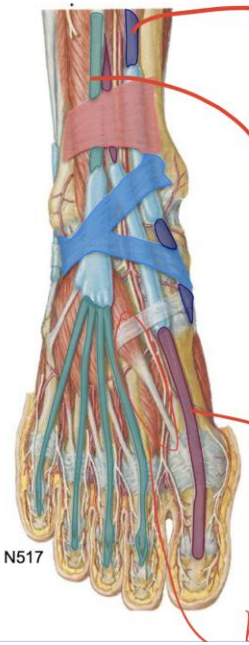
B. Sesamoid Bones – These bones sometimes form in tendons where the tendon rubs against bone. The patella is an example of a large sesamoid bone that formed in the tendon of the quadriceps as this tendon passes over the knee. In the foot, the tendons of flexor hallucis brevis will often form sesamoid bones as they pass over the 1st metatarsophalangeal joint.

Soft Tissue: Dorsum of the Foot

Extensor Retinacula

Superior Extensor Retinaculum

Inferior Extensor Retinaculum



tibialis anterior

- Broad attachment on medial aspect of foot

Extensor Digitorum Longus

- Splits into 4 tendons and attaches to extensor expansion on dorsal digits
- Functions with Extensor Digitorum Brevis

big toe

Extensor Hallucis Longus

- Extends to extensor expansion of 1st digit
- Functions with Extensor Hallucis Brevis

brevis

III. Fascia of the Foot and Ankle

A. **Anterior Compartment:** All tendons from the anterior compartments pass under a thickening of fascia, the extensor retinacula, that bind the tendons before and after they cross the ankle joint preventing them from bowstringing anteriorly during dorsiflexion. The are two retinacula:
   a. Two bind tendons of the **anterior compartment**:
      i. **Superior Extensor Retinaculum** – travels from the tibia to **the fibula anteriorly**
      ii. **Inferior Extensor Retinaculum** – Y-shaped with stem arising from calcaneus, the superior branch going to the medial malleolus, and the inferior branch going to deep fascia.

Soft Tissue: Sole of the Foot

- Similar to Palm of Hand
- **Plantar Fascia**
  - Thick Central Part is the Plantar Aponeurosis
    - Protects Sole
    - Supports Longitudinal Arches
    - Holds Everything Together
- Four Layers of Muscles
  - Function as a Group to Support Stance and Maintain Arches





Sole of Foot: The deep fascia of the lower limb is continuous with the **plantar fascia** on the sole of the foot, which has a thick central part known as the **plantar aponeurosis.**

Clinical Note: Plantar Fasciitis – inflammation of the **plantar fascia** caused by an overuse mechanism (i.e. running, aerobics, etc.), often from improper footwear.

- Causes pain on the plantar aspect of the heel and foot, especially the medial aspect of the **plantar aponeurosis**.
- Pain increases when the hallux is extended and is further exacerbated with dorsiflexion of the ankle.

Tendons: Sole of the Foot



Flexor Digitorum Longus

- Splits into 4 tendons, passes through a split in Flexor Digitorum Brevis tendons to distal phalanges of digits 2-5

Quadratus Plantae

- Assists Flexor Digitorum Longus by changing the Line of Pull

Flexor Hallucis Longus

- Passes through 2 heads of Flexor Hallucis Brevis and sesamoid bones to distal phalanx of 1st digit



Posterior Compartment: the **flexor retinaculum** is a transverse thickening of deep fascia that holds the tendons of the deep posterior compartment muscles as they cross the ankle joint *posterior to the medial malleolus*

IV. Muscles of the Sole of the Foot

a. Unlike the hand, the muscles of the plantar foot are of little importance individually because fine movements of the foot are not important to most individuals. They function primarily as a group during the support phase of stance to maintain the arches of the foot. These muscles resist forces that tend to reduce the longitudinal arch as weight is transferred from the heel to the ball of the foot.
b. These muscles are most active, stabilizing the foot, during propulsion (toe-off). They also function to help the long muscles refine movements, such as pronation and supination, enabling the foot to adjust to uneven ground.
c. These muscles can be categorized in two different ways. The first way is to divide the muscles into 4 layers (from superficial to deep). The other way is to divide the muscles into medial, intermediate, lateral and interosseous groups. The 4th muscle layer and the interosseous group are synonymous.

1. Muscle Groups based on Layers

a. **First Layer:** abductor hallucis, flexor digitorum brevis, and abductor digiti minimi
b. **Second Layer:** quadratus plantae and lumbricals
c. **Third Layer:** flexor hallucis brevis, adductor hallucis, flexor digiti minimi brevis
d. **Fourth Layer:** plantar interossei (3) and dorsal interossei (4)

2. Muscle Groups based on Compartments

a. **Medial Compartment:** abductor hallucis, flexor hallucis brevis
b. **Intermediate Compartment:** flexor digitorum brevis, quadratus plantae, lumbricals, adductor hallucis
c. **Lateral Compartment:** flexor digiti minimi and abductor digiti minimi
d. **Interosseous Compartment:** plantar interossei (3) and dorsal interossei (4)

Ankle & Foot Joints

- Ligaments of the Ankle
- Joints and Ligaments of the Foot

Talocrural (Ankle) Joint

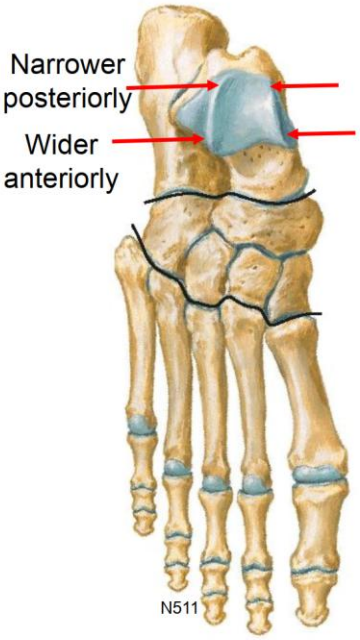
Hinge Synovial Joint

Articulation:
- Distal ends of tibia and fibula (crura) and superior aspect (trochlea) of talus

Movements:
- Hinge motion (plantar/dorsi-flexion)
- Note: congruency is highest during dorsiflexion and lowest during plantarflexion





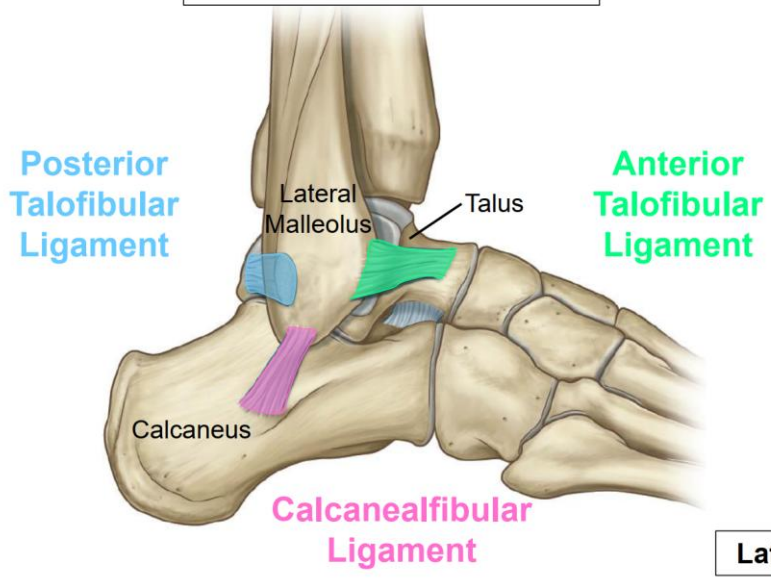


V. Ankle Joint (Talocrural Joint)

a. **Type:** hinge synovial joint
b. **Articulation:** distal ends of tibia and fibula articulate with the trochlea of the talus
   i. Fibular malleolus (sharp) extends ~2 cm lower than tibial malleolus (blunt)
   ii. **Congruency is highest** during **dorsiflexion** of foot (i.e. most stable when 'digging in one's heels'). In dorsiflexion the talus is the widest, resulting in the anterior part of trochlea being wedged between the malleoli
   iii. C**ongruency is lowest** during p**lantarflexion** of the foot (i.e. least stable position) when the narrower, posterior part of trochlea is placed between the malleoli there is a looser fit.

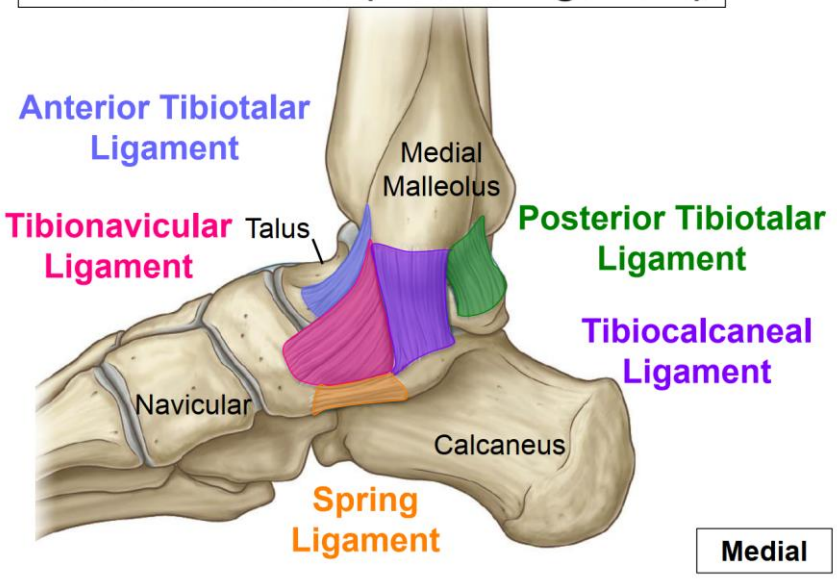
Ankle Joint: Ligaments

Lateral Collateral



Ankle Joint: Ligaments

Medial Collateral (Deltoid Ligament)



C. Supporting Structures: Joint capsule is thin/weak anteriorly and posteriorly (facilitating movement), but is reinforced medially and laterally by collateral ligaments

1. **Lateral collateral ligament:** relatively weak, offers minor protection against inversion of the foot; consists of three bands fanning out from fibula to the talus and the calcaneus:
    a. Anterior talofibular (ATF) ligament
    b. Posterior talofibular (PTF) ligament
    c. Calcaneofibular (CF) ligament
    d. **Clinical Note:** The ATF is the most vulnerable and commonly torn/damaged during inversion ankle sprains
2. **Medial collateral ligament (deltoid ligament):** thick and strong; stabilizes ankle during eversion; consisting of four bands extending from the medial malleolus of the tibia to the talus, the calcaneus, and the navicular bones:
    a. Anterior tibiotalar (ATT) ligament
    b. Posterior tibiotalar (PTT) ligament
    c. Tibiocalcaneal (TC) ligament
    d. Tibionavicular (TN) ligament
    e. **Clinical Note:** damaged collateral ligaments (ankle sprains) are more likely to occur laterally (esp. weak anterior talofibular ligament) due to the tendency for the ankle to invert. Severe inversion/eversion events may also result in avulsion bone fractures known as **Bimalleolar Fracture-Dislocations** of the ankle. In the ER, X-rays for suspected sprained ankles are called for based on the **Ottawa Ankle Rules:**
        i. If there is **bone tenderness** along posterior aspect of **distal tibia** or tip of **medial malleolu**s; OR
        ii. If there is **bone tenderness** along posterior aspect of **distal fibula** or tip of **lateral malleolus**; OR
        iii. If the patient is **unable to bear weight** for **four steps** both immediately and in the ER

D. Movements:

1. **Dorsiflexion** - by tibialis anterior, fibularis tertius, and extensor hallucis longus and extensor digitorum longus.
2. **Plantarflexion** - gastrocnemius and soleus via calcaneal (Achilles) tendon of foot, with some wobble (small degrees of adduction/abduction, inversion/eversion) allowed in unstable plantarflexed position

Subtalar (Talocalcaneal) Joint

Plane Synovial Joint

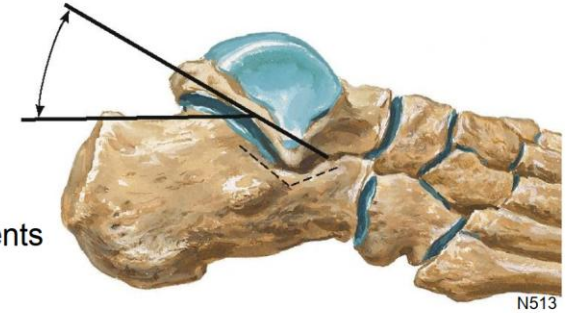
Articulation:
- Inferior surface of talus, superior aspect of calcaneus

Movements:
- Bulk of inversion/eversion motion occurs here

Support:
- Interosseous talocalcaneal ligaments





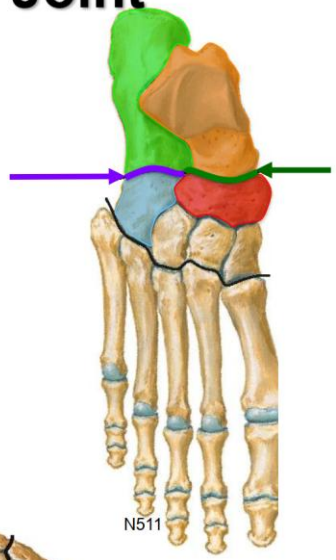


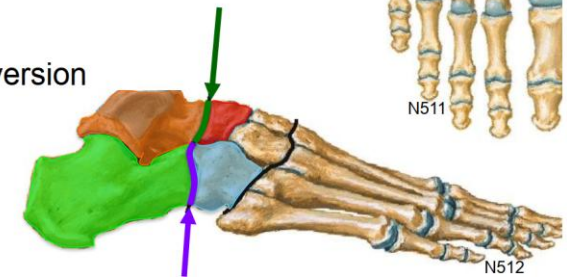
Transverse Tarsal Joint

Creates **subtalor joint complex** with Talocalcaneal Joint

Compound of TWO Synovial Joints:
- **Calcaneocuboid** (plane)
  - **Calcaneus** with **cuboid** laterally
- **Talonavicular part** of talocalcaneonavicular joint (ball and socket)
  - **Head of talus** with **navicular** medially

Movements:
- Assists with inversion/eversion





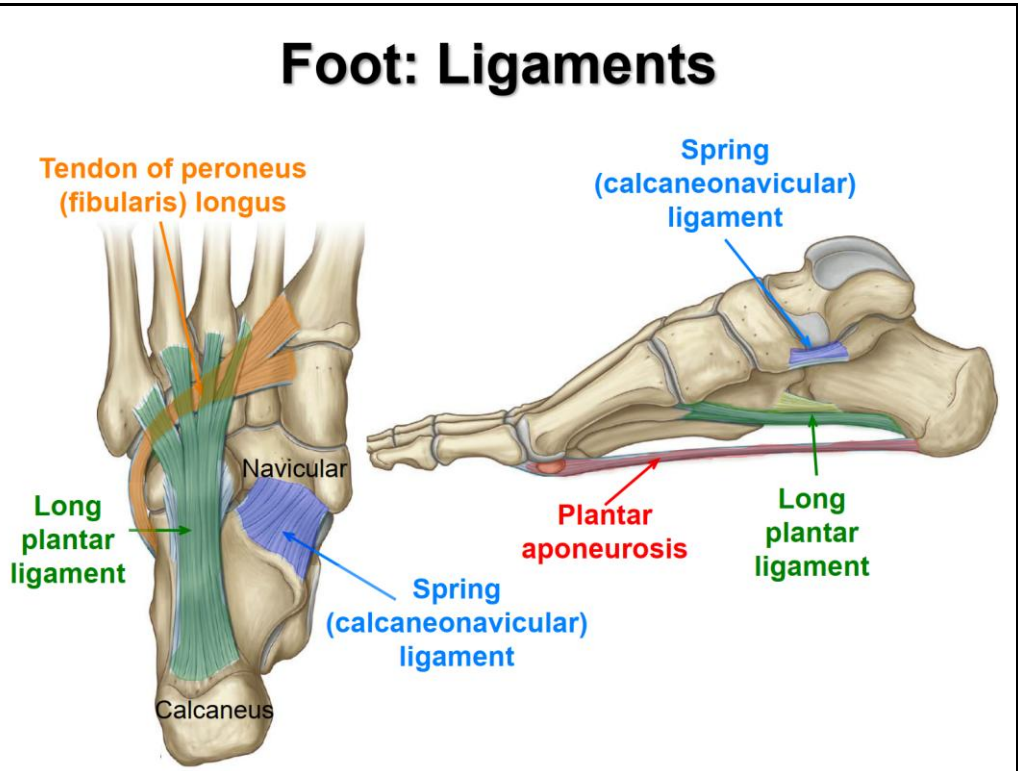
VI. Joints of Foot

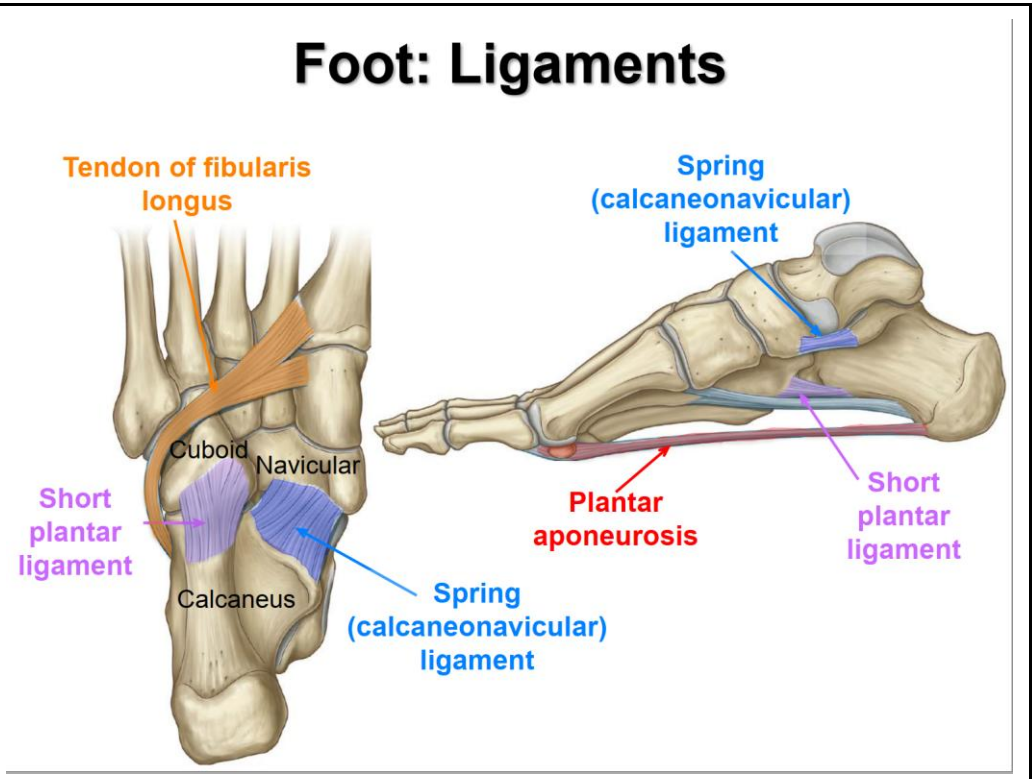
A. Subtalar Joint (talocalcaneal joint)

a. **Type:** plane synovial joint
b. **Articulation:** inferior surface of talus; superior surface of calcaneus
c. **Supporting Structures:** Joint surrounded by weak joint capsule supported by interosseous talocalcaneal ligaments
d. **Movements:** The bulk of foot inversion/eversion happens at this joint
   i. **Inverters:** tibialis anterior and posterior
   ii. **Everters:** fibularis longus and brevis (primary purpose: resist inversion injuries)
e. **Clinical Note: Calcaneal Fractures** - A calcaneal fracture can occur when a person falls hard on their heel. Often the calcaneus will break into a number of pieces. Calcaneal fractures are usually disabling because it disrupts the subtalar joint.

B. Transverse tarsal joint

a. Compound of two synovial joints, each joint has its own joint cavities but they are aligned transversely and thus function as a physiologic unit:
   i. **Talonavicular part** of the **talocalcaneonavicular joint**
      1. **Type:** Ball and socket synovial joint
      2. **Articulation:** Talus and navicular bones
   ii. **Calcaneocuboid**
      1. **Type:** Plane synovial joint
      2. **Articulation:** Calcaneus and cuboid bones



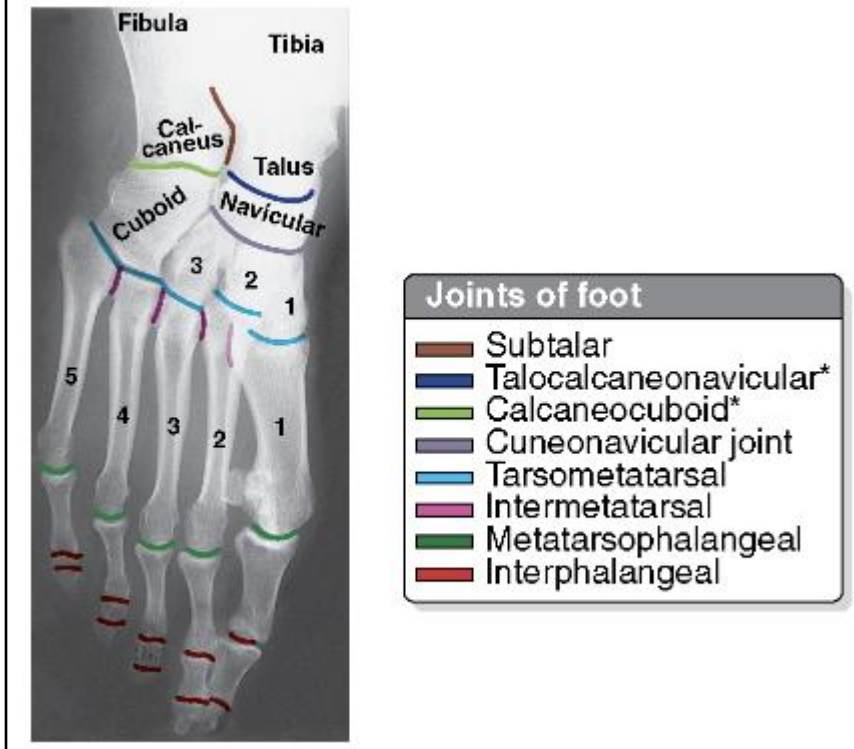


2. Joint is supported on plantar surface by several ligaments:

a. **Plantar calcaneonavicular (spring) ligament:** extends across and fills a wedge-shaped gap between the sustentaculum tali of the calcaneus and the navicular.
   i. This ligament supports the head of the talus and plays an important role in the transfer of weight from the talus.
   ii. It is also considered to be the 'keystone' along with the head of the talus in maintaining the longitudinal arch of the foot.
   iii. **Clinical Note:** if the spring ligament ruptures or is stretched, the head of the talus may protrude between the calcaneus and the navicular, flattening the medial arch and forcing the anterior foot into abduction/eversion

b. **Long plantar ligament:** extends from the plantar aspect of the calcaneus to the cuboid.
   i. Some of its fibers extend to the bases of the metatarsals forming a tunnel for the tendon of fibularis longus.
   ii. Important in maintaining the longitudinal arch of the foot.

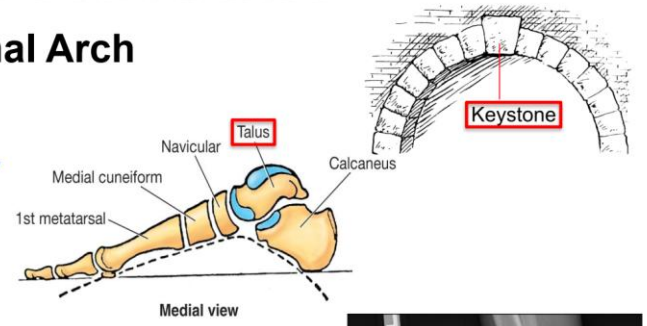
c. **Short plantar (plantar Calcaneocuboid) ligament:** extends from the calcaneus to the cuboid in a plane between the long plantar ligament and the spring ligament.
   i. Involved with maintaining the **longitudinal arch** of the foot.


Fibula
Tibia
Cal-
caneus
Talus
Cuboid
Navicular
3
2
1
5
4
3
2
1
Joints of foot
Subtalar
Talocalcaneonavicular*
Calcaneocuboid*
Cuneonavicular joint
Tarsometatarsal
Intermetatarsal
Metatarsophalangeal
Interphalangeal

Foot: Arches

Medial Longitudinal Arch

- Supported by:
  - Tibialis Anterior
  - Tibialis Posterior
  - Peroneus Longus
  - Foot Ligaments



Medial view

Unhappy Feet:

Pes Planus ("Flat Feet")

- Often caused by damage to **Tibialis Posterior**
  - Results in a dropped talus
- Orthotics if Symptomatic

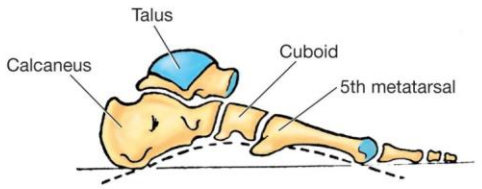




Foot: Arches

Lateral Longitudinal Arch

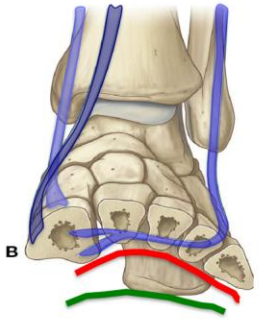
- Much flatter than medial
- Rests on the ground
- Supported by foot ligaments



Lateral view

Transverse Arch

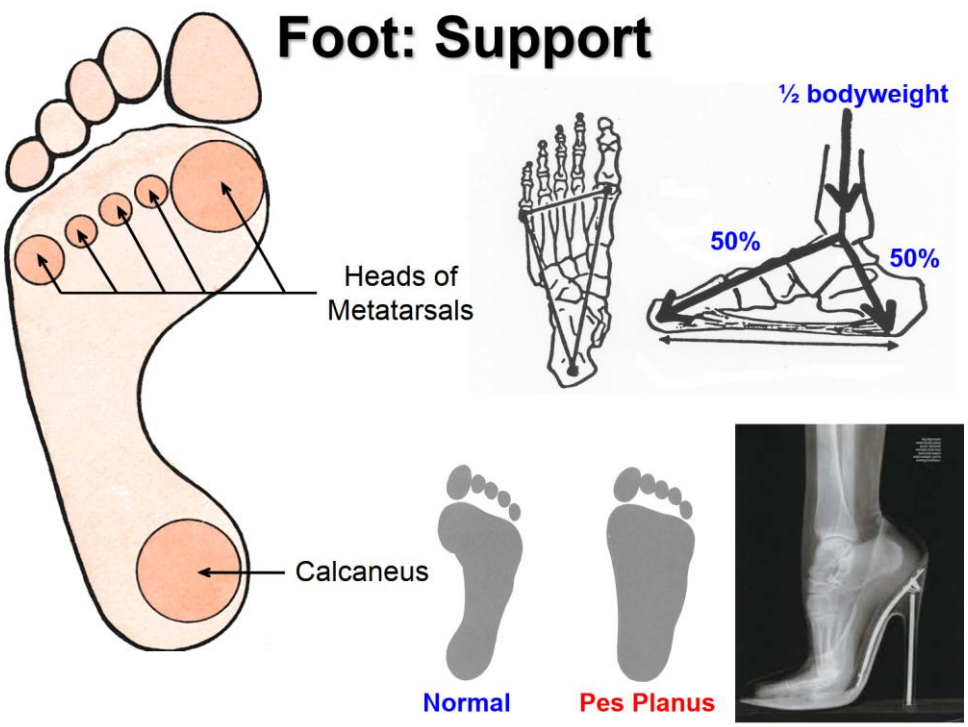
- Cuboid, cuneiforms, metatarsals
- Runs from medial to lateral
- Pillars formed by medial and lateral longitudinal arches
- Supported by:
  - Tibialis Anterior
  - Tibialis Posterior
  - **Peroneus Longus**



VII. Arches of the Foot

The arches of the foot distribute weight over the foot acting as shock absorbers and spring-boards (for walking, running and jumping). Weight is transmitted from the tibia to the talus and then to the calcaneus posteriorly and the sesamoid bones of the 1st metatarsal and the metatarsal heads of digits 2-5 anteriorly. Between the weight-bearing points the elastic arches of the foot flatten and rebound when weight is applied and removed.

1. **The Longitudinal Arch of the Foot** is composed of two parts (the medial and lateral longitudinal arches) that functionally act as a unit with the transverse arch of the foot spreading weight in all directions.
2. **The Medial Longitudinal Arch:** which is higher and more prominent than the lateral arch.
   a. Formed by the calcaneus, talus, navicular, three cuneiforms, and three metatarsals.
   b. The talar head is the 'keystone' of the medial longitudinal arch.
   c. The tibialis anterior and posterior and the fibularis longus tendon help to support this arch.
3. **The Lateral Longitudinal Arch:** much flatter and rests on the ground while standing.
   a. Formed by the calcaneus, cuboid, and lateral two metatarsals.
4. **The Transverse Arch of the Foot:** runs from medial to lateral.
   a. Formed by the cuboid, cuneiforms, and the bases of the metatarsals.
   b. The fibularis longus and tibialis posterior maintain the curvature of the transverse arch.

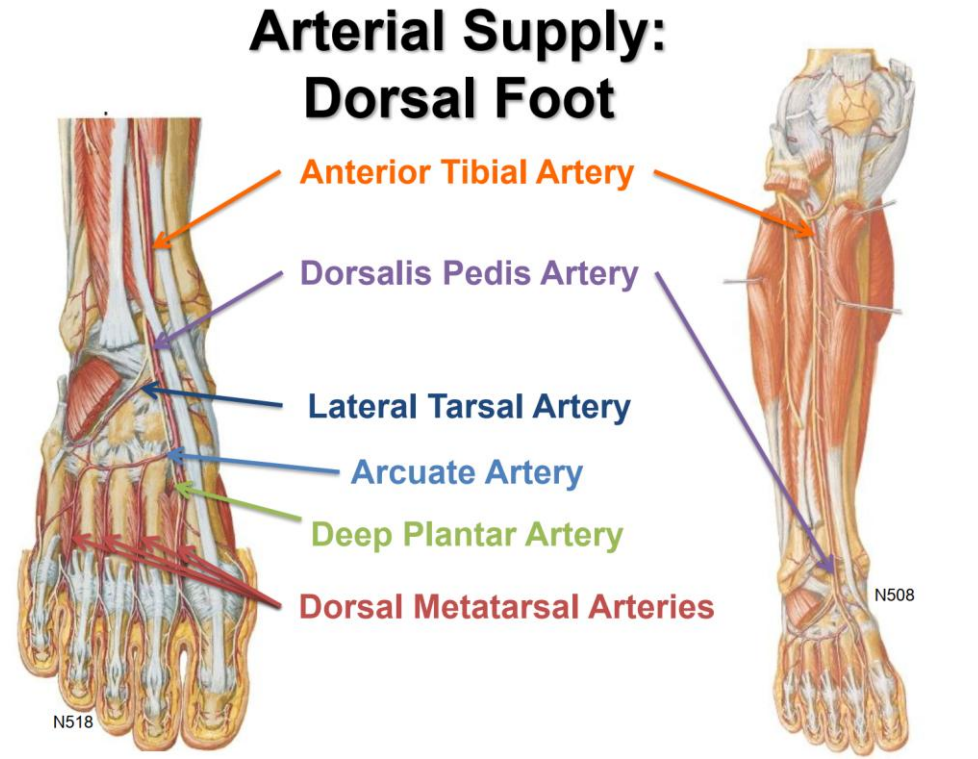


Clinical Note: Pes Planus (aka flat feet): Acquired pes planus often occurs after an injury to or degeneration of tibialis posterior. Without the support of the tibialis posterior tendon, the talar head displaces inferiorly and medially. This flattens the **medial longitudinal arch.**

- Though this condition is most often asymptomatic, if foot or lower leg pain accompanies, treatment can help. Shoes with orthotics (arch support) and/or foot exercises can relieve symptoms.

Neurovascular Structures of the Foot

- Arterial Supply
- Nerve Supply



you

Dorsalis Pedis Pulse

- Diminished or absent Dorsalis Pedis Pulse usually indicative of vascular insufficiency
  - Resulting from Arterial Disease



- Five P's of Arterial Occlusion
  - **P**ain
  - **P**allor
  - **P**aresthesia
  - **P**aralysis
  - **P**ulselessness

VIII. Vascular Structures of the Foot

a. **Review**: Aorta → Common Iliac Artery → External Iliac Artery → Femoral Artery → Popliteal Artery → Anterior and Posterior Tibial Arteries

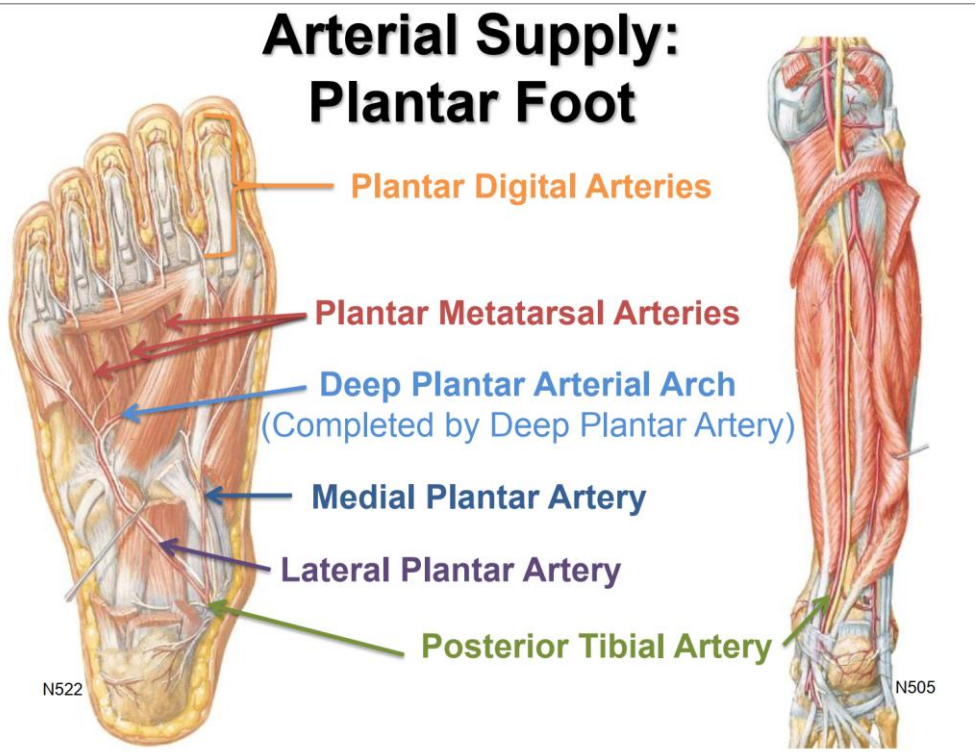
b. **Dorsalis Pedis** – terminal branch of the **anterior tibial artery** starting between the malleoli. The dorsalis pedis courses on the dorsum of the foot giving off the lateral tarsal artery and **arcuate artery** before giving off the **deep plantar artery** and **1st dorsal metatarsal artery** as terminal branches.

i. **Arcuate Artery** – a branch of dorsalis pedis. The arcuate artery comes off at the base of the 2nd metatarsal and courses laterally along the dorsal side of the metatarsals before anastomosing with the lateral tarsal artery. The arcuate artery gives rise to the 2nd, 3rd, and 4th dorsal metatarsal arteries.

ii. **Deep Plantar Artery** – branch of the dorsalis pedis. The deep plantar artery dives through the space between the 1st and 2nd metatarsals to reach the plantar foot, where it joins with the **lateral plantar artery,** a branch of the posterior tibial artery, to form the **deep plantar arch.**

iii. **Clinical Note: Dorsalis Pedis Pulse:** The dorsalis pedis pulse is evaluated on physical examination of the vascular system. The pulse is easily detected because of the superficial nature of the dorsalis pedis artery on the foot. A weak or absent pulse can be indicative of vascular insufficiency. Acute arterial occlusion can be assessed using the **five P's**:

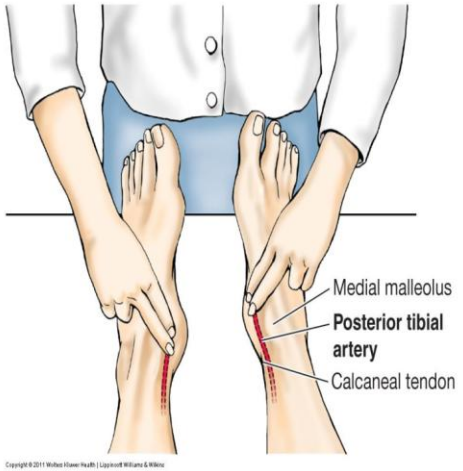
1. Pain
2. Pallor
3. Paresthesia
4. Paralysis
5. Pulselessness



Posterior Tibial Pulse

- Checking pulse at or near the tarsal tunnel.
- Between medial malleolus and calcaneal tendon.
- Invert foot to relax flexor retinaculum.
- Both pulses simultaneously

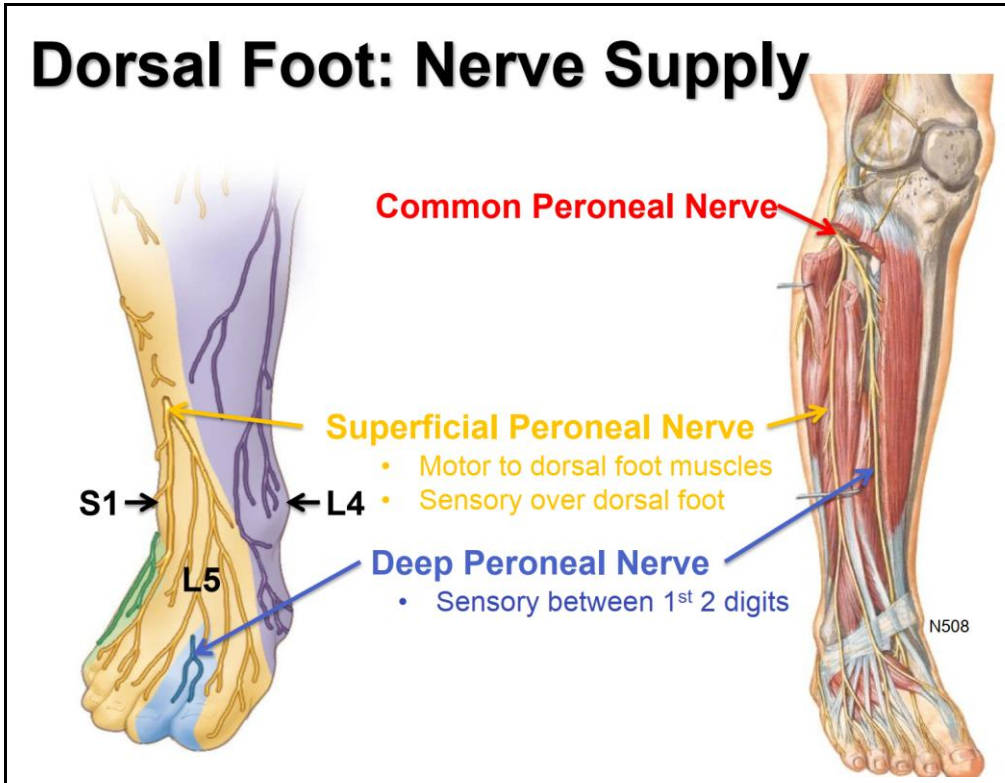
Symmetry!!!



C. Posterior Tibial Artery passes through the **tarsal tunnel** and then divides into the **lateral plantar artery** and the **medial plantar artery**, which travel through the 1st and 2nd muscle layers.

1. **Lateral Plantar Artery** runs along the lateral aspect of the sole. It then arches medially across the foot to form the **deep plantar arch**, which is completed by the deep plantar artery (branch of dorsalis pedis artery).
    a. The **deep plantar arch** gives rise to four **plantar metatarsal arteries.**
        i. **Plantar metatarsal arteries** divide near the base of the proximal phalanges to form the **plantar digital arteries.**
2. **Medial Plantar Artery** travels along the medial side of the foot where it contributes to the **deep plantar arterial arch** and branches to supply the medial sole and great toe.
    a. **Clinical Note: Wounds to the sole of the foot:** puncture/perforating wounds to the sole of the foot are relatively common. Any puncture wound involving the deep plantar arch and its branches usually results in severe bleeding, typically from both ends of the cut artery because of the abundant anastomoses. Additionally, this is a difficult artery to ligate due to its depth and surrounding structures.
    b. **Clinical Note: Diabetic foot ulcers:** patients with diabetes mellitus commonly have ulcers form on the soles of their feet. These are caused by many factors such as changes in bony architecture and atherosclerotic peripheral artery disease. Due to poor circulation and extracellular matrix problems, these wounds often do not heal on their own. Additionally, these patients often go untreated because they have peripheral neuropathy (or malfunction of distal nerves, especially in the hands and feet), meaning they are unable to feel the ulcer, and being on the bottom of their foot, they simply do not notice it. These ulcers may require very specialized treatment, surgery, and sometimes amputation if not controlled.

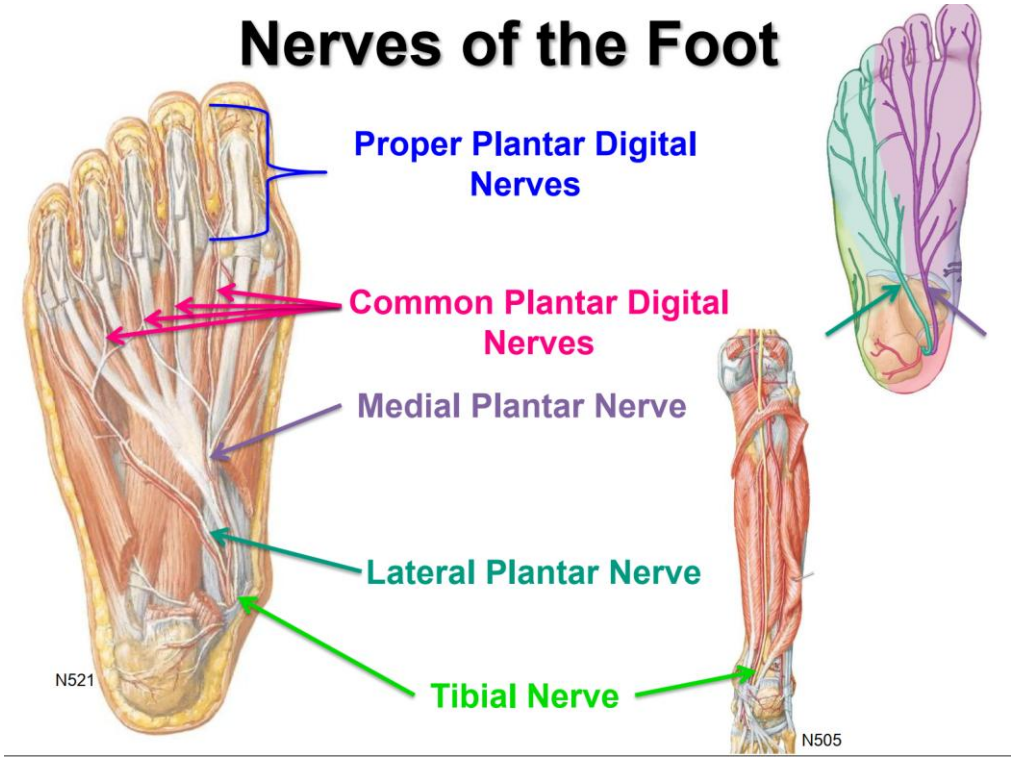
D. Venous drainage: Plantar venous blood drains from a ***plantar venous network*** around the medial or lateral side of the foot to join up with the ***dorsal venous network.*** From there it leaves the foot via the *great saphenous vein* and the *small saphenous vein.*



IX. Nervous Supply to the Foot

A. **Superficial Fibular Nerve** courses from the lateral compartment to the dorsum of the foot where it supplies sensation to the entire dorsum, not including the space between the first two digits.

B. **Deep Fibular Nerve** courses with dorsalis pedis artery to enter the dorsum of the foot to innervate the muscles of the dorsal foot. Finally, the deep fibular nerve gives cutaneous innervation to the space between the big and 2nd toe.



C. Tibial Nerve courses down the posterior leg and through the tarsal tunnel. On the sole of the foot it splits into the medial and lateral plantar nerves. These nerves course through the 1st and 2nd layer muscles.

1. **Medial Plantar Nerve** supplies three medial plantar foot muscles and then terminates as three sensory branches, **common plantar digital nerves**, which will branch into p**roper plantar digital nerves** that provide sensory innervation to the medial three and half digits.
2. **Lateral Plantar Nerve** gives off the ***deep branch of the lateral plantar nerve*** and continues as the ***superficial branch of the lateral plantar nerve***, which splits into sensory branches, ***common & proper plantar digital nerves,*** innervating the lateral one and half digits. The s***uperficial and deep branches of the lateral plantar nerve*** supply all the muscles of the sole not supplied by the ***medial plantar nerve.***

Note: Lymphatic drainage: The lymphatic drainage in the foot follows venous drainage. The deep lymphatic vessels drain mostly into the popliteal lymph nodes.